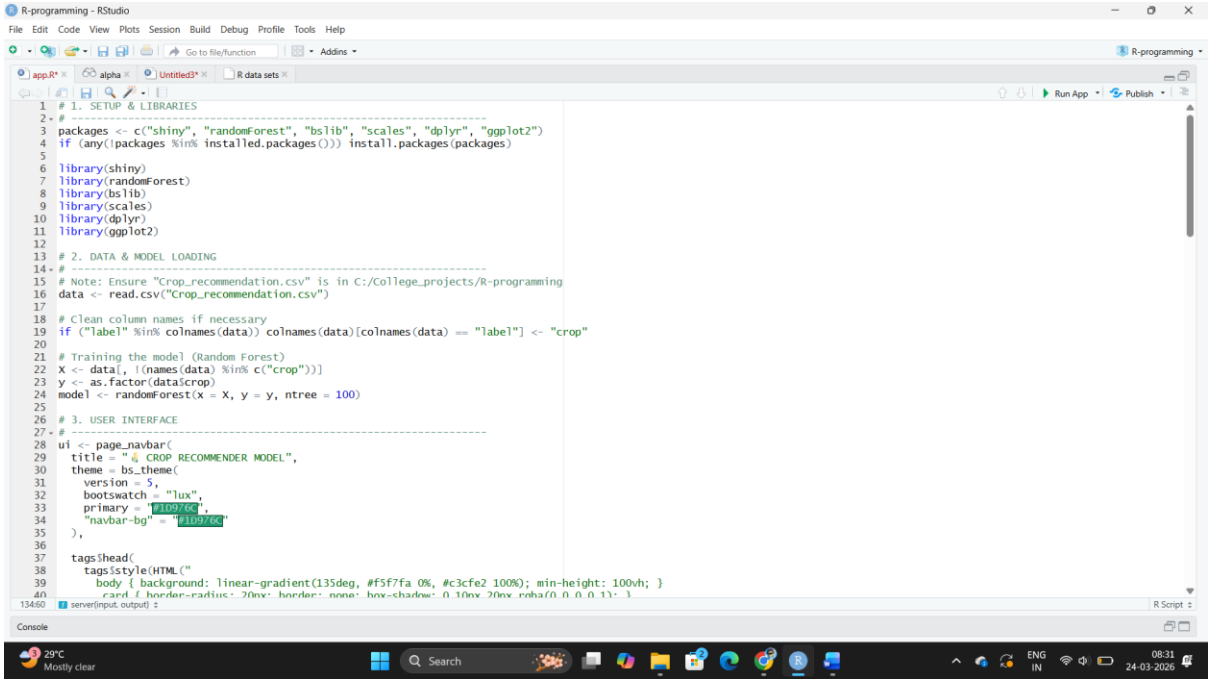
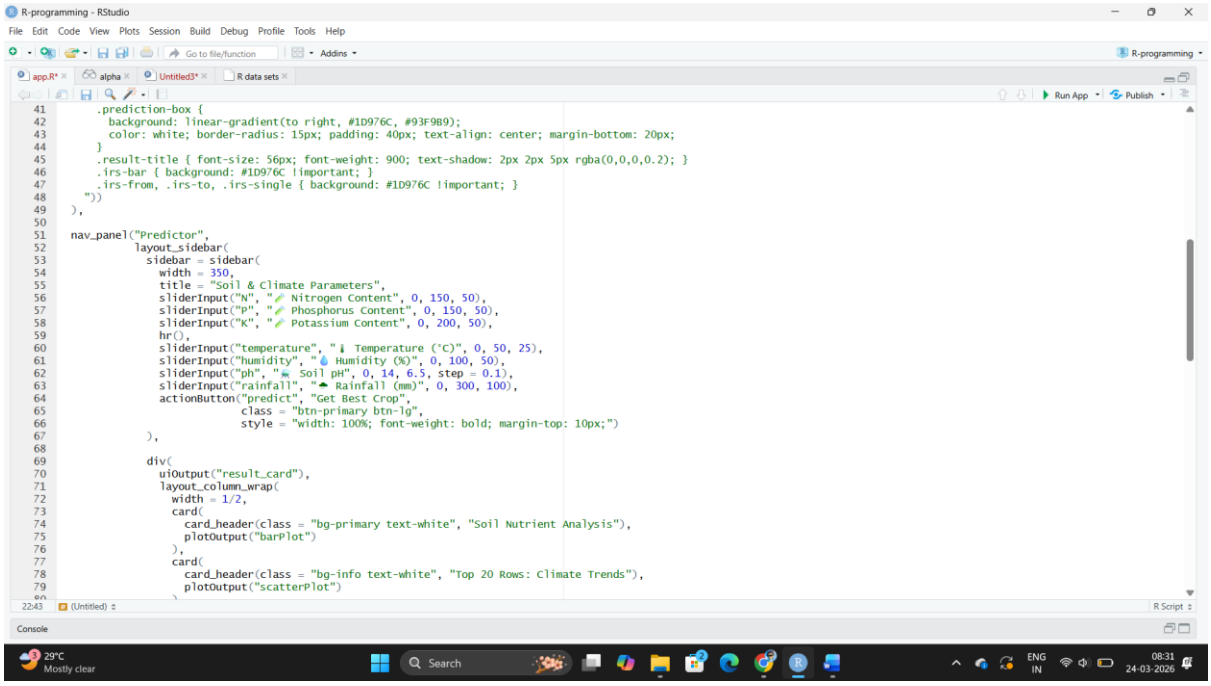
SHETH LUJ COLLEGE

R - PROJECT



```r
# 1. SETUP & LIBRARIES
# ----------------------------------------------------------------
packages <- c("shiny", "randomForest", "bslib", "scales", "dplyr", "ggplot2")
if (any(!packages %in% installed.packages())) install.packages(packages)

library(shiny)
library(randomForest)
library(bslib)
library(scales)
library(dplyr)
library(ggplot2)

# 2. DATA & MODEL LOADING
# ----------------------------------------------------------------
# Note: Ensure "Crop_recommendation.csv" is in C:/College_projects/R-programming
data <- read.csv("Crop_recommendation.csv")

# Clean column names if necessary
if ("label" %in% colnames(data)) colnames(data)[colnames(data) == "label"] <- "crop"

# Training the model (Random Forest)
X <- data[, !(names(data) %in% c("crop"))]
y <- as.factor(data$crop)
model <- randomForest(x = X, y = y, ntree = 100)

# 3. USER INTERFACE
# ----------------------------------------------------------------
ui <- page_navbar(
  title = "🌾 CROP RECOMMENDER MODEL",
  theme = bs_theme(
    version = 5,
    bootswatch = "lux",
    primary = "#1D976C",
    "navbar-bg" = "#1D976C"
  ),

  tags$head(
    tags$style(HTML("
      body { background: linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%); min-height: 100vh; }
      .card { border-radius: 20px; border: none; box-shadow: 0 10px 20px rgba(0,0,0,1); }
```



```r
      .prediction-box {
        background: linear-gradient(to right, #1D976C, #93F9B9);
        color: white; border-radius: 15px; padding: 40px; text-align: center; margin-bottom: 20px;
      }
      .result-title { font-size: 56px; font-weight: 900; text-shadow: 2px 2px 5px rgba(0,0,0,0.2); }
      .irs-bar { background: #1D976C !important; }
      .irs-from, .irs-to, .irs-single { background: #1D976C !important; }
    ."))
  ),

  nav_panel("Predictor",
            layout_sidebar(
              sidebar = sidebar(
                width = 350,
                title = "Soil & Climate Parameters",
                sliderInput("N", "🌱 Nitrogen Content", 0, 150, 50),
                sliderInput("P", "🌱 Phosphorus Content", 0, 150, 50),
                sliderInput("K", "🌱 Potassium Content", 0, 200, 50),
                hr(),
                sliderInput("temperature", "🌡 Temperature (°C)", 0, 50, 25),
                sliderInput("humidity", "💧 Humidity (%)", 0, 100, 50),
                sliderInput("ph", "🧪 Soil pH", 0, 14, 6.5, step = 0.1),
                sliderInput("rainfall", "☔ Rainfall (mm)", 0, 300, 100),
                actionButton("predict", "Get Best Crop",
                             class = "btn-primary btn-lg",
                             style = "width: 100%; font-weight: bold; margin-top: 10px;")
              ),

              div(
                uiOutput("result_card"),
                layout_column_wrap(
                  width = 1/2,
                  card(
                    card_header(class = "bg-primary text-white", "Soil Nutrient Analysis"),
                    plotOutput("barPlot")
                  ),
                  card(
                    card_header(class = "bg-info text-white", "Top 20 Rows: Climate Trends"),
                    plotOutput("scatterPlot")
```

DANIYAL KHAN
S088

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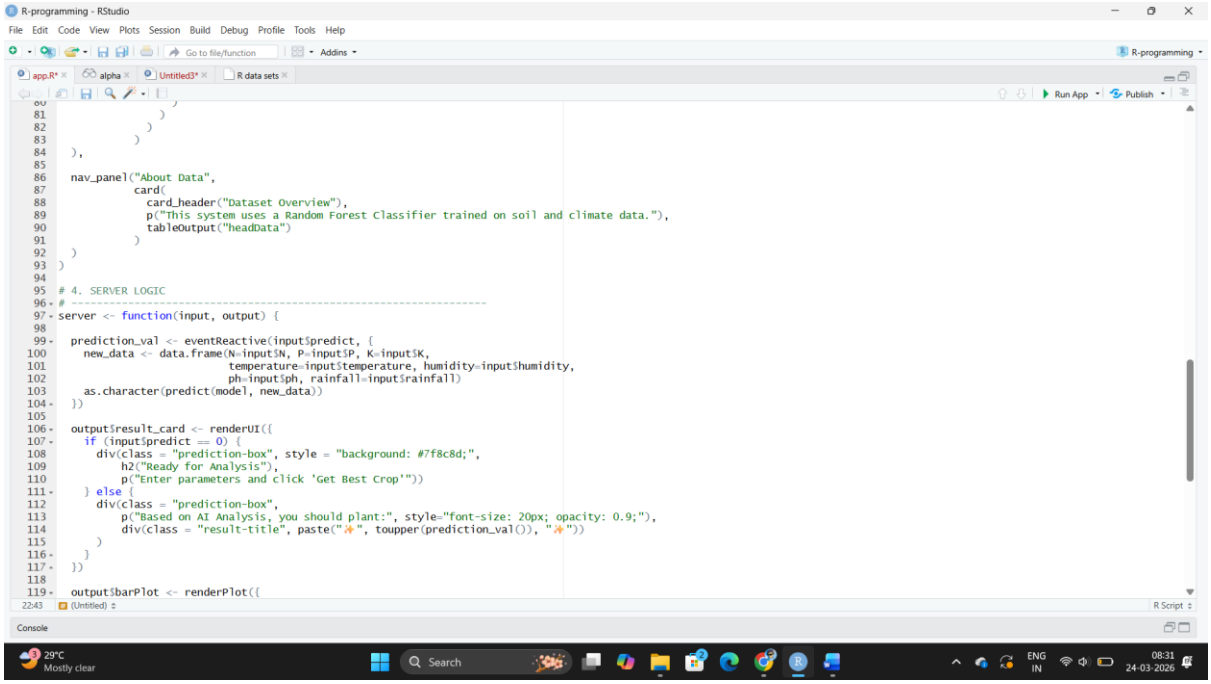
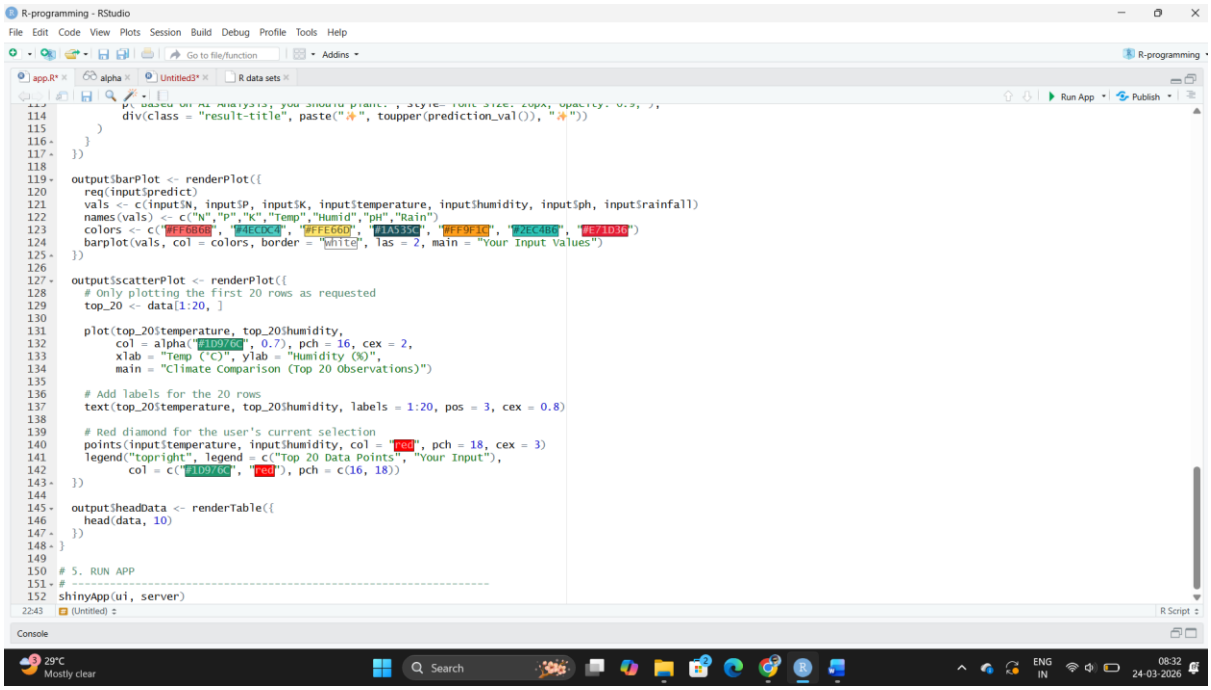


```
                )
              )
            )
  ),

  nav_panel("About Data",
            card(
              card_header("Dataset Overview"),
              p("This system uses a Random Forest Classifier trained on soil and climate data."),
              tableOutput("headData")
            )
  )
)

# 4. SERVER LOGIC
# ----------------------------------------------------------------
server <- function(input, output) {

  prediction_val <- eventReactive(input$predict, {
    new_data <- data.frame(N=input$N, P=input$P, K=input$K,
                           temperature=input$temperature, humidity=input$humidity,
                           ph=input$ph, rainfall=input$rainfall)
    as.character(predict(model, new_data))
  })

  output$result_card <- renderUI({
    if (input$predict == 0) {
      div(class = "prediction-box", style = "background: #7f8c8d;",
          h2("Ready for Analysis"),
          p("Enter parameters and click 'Get Best Crop'"))
    } else {
      div(class = "prediction-box",
          p("Based on AI Analysis, you should plant:", style="font-size: 20px; opacity: 0.9;"),
          div(class = "result-title", paste("✨", toupper(prediction_val()), "✨"))
      )
    }
  })

  output$barPlot <- renderPlot({
```



```
          div(class = "result-title", paste("✨", toupper(prediction_val()), "✨"))
      )
    }
  })

  output$barPlot <- renderPlot({
    req(input$predict)
    vals <- c(input$N, input$P, input$K, input$temperature, input$humidity, input$ph, input$rainfall)
    names(vals) <- c("N","P","K","Temp","Humid","pH","Rain")
    colors <- c("#FF6B6B", "#4ECDC4", "#FFE66D", "#1A535C", "#FF9F1C", "#2EC4B6", "#E71D36")
    barplot(vals, col = colors, border = "white", las = 2, main = "Your Input Values")
  })

  output$scatterPlot <- renderPlot({
    # Only plotting the first 20 rows as requested
    top_20 <- data[1:20, ]

    plot(top_20$temperature, top_20$humidity,
         col = alpha("#1D976C", 0.7), pch = 16, cex = 2,
         xlab = "Temp (°C)", ylab = "Humidity (%)",
         main = "Climate Comparison (Top 20 Observations)")

    # Add labels for the 20 rows
    text(top_20$temperature, top_20$humidity, labels = 1:20, pos = 3, cex = 0.8)

    # Red diamond for the user's current selection
    points(input$temperature, input$humidity, col = "red", pch = 18, cex = 3)
    legend("topright", legend = c("Top 20 Data Points", "Your Input"),
           col = c("#1D976C", "red"), pch = c(16, 18))
  })

  output$headData <- renderTable({
    head(data, 10)
  })
}

# 5. RUN APP
# ----------------------------------------------------------------
shinyApp(ui, server)
```

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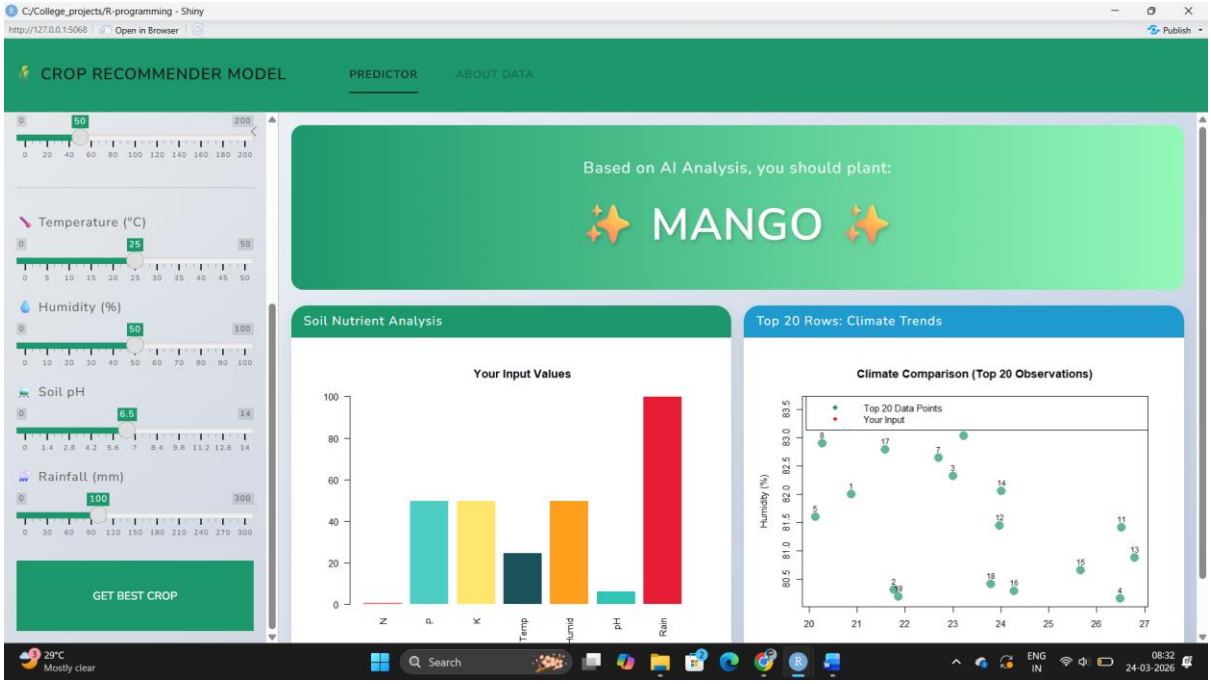
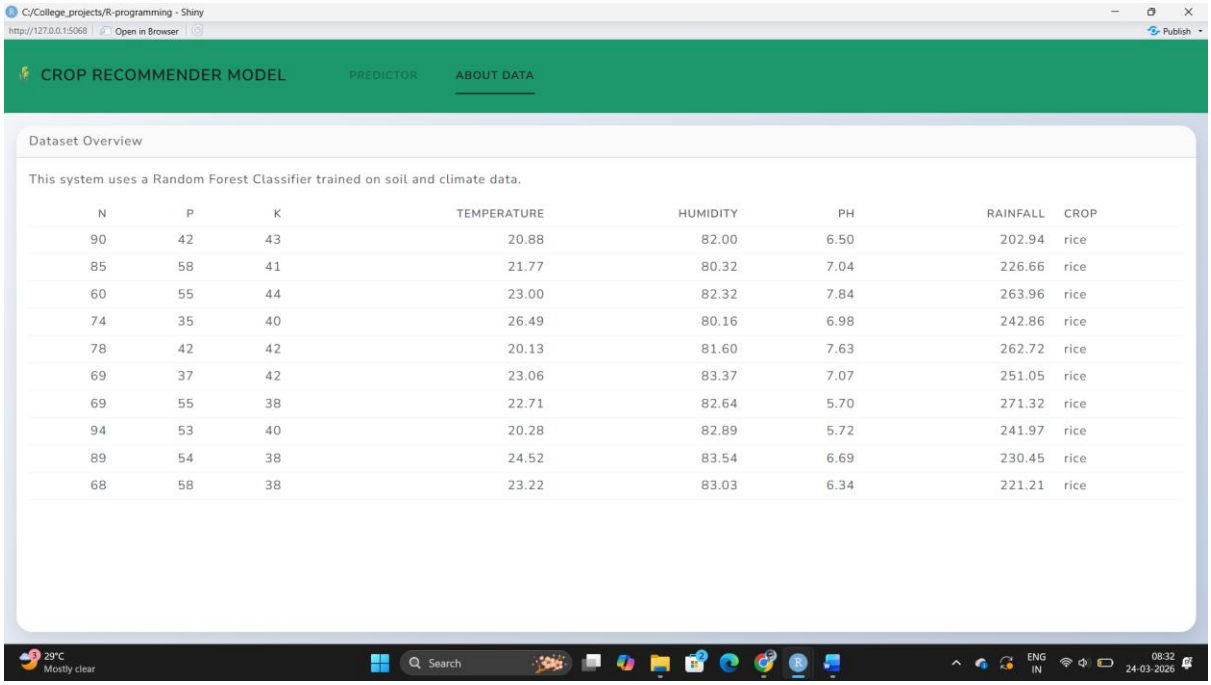
CODE:-





N	P	K	TEMPERATURE	HUMIDITY	PH	RAINFALL	CROP
90	42	43	20.88	82.00	6.50	202.94	rice
85	58	41	21.77	80.32	7.04	226.66	rice
60	55	44	23.00	82.32	7.84	263.96	rice
74	35	40	26.49	80.16	6.98	242.86	rice
78	42	42	20.13	81.60	7.63	262.72	rice
69	37	42	23.06	83.37	7.07	251.05	rice
69	55	38	22.71	82.64	5.70	271.32	rice
94	53	40	20.28	82.89	5.72	241.97	rice
89	54	38	24.52	83.54	6.69	230.45	rice
68	58	38	23.22	83.03	6.34	221.21	rice

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